

Psyche's Martian Slingshot: NASA Probe to Skim Mars at 12,333 MPH in Daring Gravity Assist

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NASA's Psyche mission is about to pull off a high-speed cosmic maneuver. On Friday, May 15, the spacecraft will scream past Mars at 12,333 miles per hour, passing just 2,800 miles above the red planet. This isn't a sightseeing detour; it's a precisely calculated **gravity assist**, a slingshot effect that will bend Psyche's trajectory and boost its velocity, saving precious propellant on its 2.2-billion-mile journey to the asteroid Psyche. Launched in October 2023, Psyche relies on a **solar-electric propulsion** system that expels **xenon gas** to generate gentle but continuous thrust. Even so, mission planners at NASA's Jet Propulsion Laboratory are leveraging Mars' gravitational pull to fine-tune the path. "We are now exactly on target for the flyby," said Sarah Bairstow, the mission's planning lead. The encounter also offers a rare opportunity to calibrate the spacecraft's instruments. As Psyche approaches from Mars' night side, its **multispectral imager** will capture the planet as a thin crescent, a stark contrast to the fully illuminated disk seen from Earth. Jim Bell, the imager lead at Arizona State University, called the sequence "both great calibration observations and just plain beautiful photos." The team will also search for a faint dusty ring around Mars, thought to be fed by micrometeorite impacts on its moons Phobos and Deimos. Beyond imaging, Psyche's **magnetometer** will study how the solar wind interacts with Mars' magnetic field, and its **gamma-ray spectrometer** will track changes in cosmic rays. Several Mars missions, including NASA's rovers and orbiters, plus ESA's Mars Express will coordinate observations, helping cross-calibrate instruments and refine navigation techniques. "The only reason for this flyby is to get a little help from Mars to speed us up," said principal investigator Lindy Elkins-Tanton. "But if we can do important testing of the science instruments, that would be the icing on the cake." The flyby is a critical milestone on the road to 2029, when Psyche will begin orbiting its namesake, a bizarre, metal-rich world that may be the exposed core of a failed planet.

Word Treasure

gravity assist -- A technique where a spacecraft uses a planet's gravitational pull to change speed and direction, like a slingshot, saving fuel.

solar-electric propulsion -- A system that uses electricity from solar panels to accelerate a gas (like xenon) and produce gentle, continuous thrust.

xenon gas -- A heavy, unreactive gas used as propellant in some spacecraft engines because it can be ionized and accelerated very efficiently.

multispectral imager -- A camera that captures pictures in different wavelengths of light - beyond what the human eye sees - to reveal details about a planet's surface.

magnetometer -- An instrument that measures magnetic fields, useful for studying a planet's interior and its interaction with the solar wind.

gamma-ray spectrometer -- A device that detects gamma rays emitted by cosmic bodies,

helping scientists figure out the composition of an asteroid or planet.

Background you need

Gravity assists have been used since NASA's Voyager missions in the 1970s - the famous 'Grand Tour' of the outer planets relied on them. By transferring a tiny bit of planetary momentum to the spacecraft, these slingshots save years of travel time and thousands of pounds of fuel.

The Psyche mission is led by principal investigator Lindy Elkins-Tanton from Arizona State University. Its target, asteroid 16 Psyche, orbits the Sun between Mars and Jupiter and is made largely of iron and nickel, unlike rocky or icy asteroids. Many scientists think it could be the metallic core left over from a planet that was smashed apart billions of years ago.

The Jet Propulsion Laboratory (JPL) in California, which manages the mission, is a world-renowned NASA center that builds and operates many interplanetary probes. The flyby will also involve coordination with ESA's Mars Express - a European orbiter that has been studying Mars since 2003 - and with NASA's rovers on the surface.

The solar-electric propulsion on Psyche uses Hall-effect thrusters, which ionize xenon gas and accelerate the ions using electric fields. This technology is much more fuel-efficient than chemical rockets but generates far less immediate power, meaning the spacecraft needs gravity assists to reach distant targets in reasonable time.

If all goes well, Psyche will enter orbit around the asteroid in 2029 and spend at least 21 months mapping its surface and studying its magnetic field - giving us our first close look at what might resemble Earth's own inaccessible core.

Break it down

LEAD: Psyche spacecraft's upcoming high-speed Mars flyby at 12,333 mph, passing just 2,800 miles above the surface.

+ TIMING: Friday, May 15.

PURPOSE: A gravity assist to bend trajectory and boost speed, saving precious propellant.

+ CONTEXT: Launched in October 2023, uses solar-electric propulsion with xenon gas.

OPPORTUNITY FOR CALIBRATION: Testing instruments during the flyby.

+ IMAGING: Multispectral imager captures Mars as a thin crescent; team will search for a dusty ring.

+ MAGNETOMETER: Studies solar wind interaction with Mars' magnetic field.

+ GAMMA-RAY SPECTROMETER: Tracks changes in cosmic rays.

COORDINATION: Multiple Mars missions (NASA rovers, orbiters, ESA's Mars Express) join observations.

MISSION MILESTONE: On track for 2029 arrival at asteroid Psyche, possibly an exposed planetary core.

+ CLOSING QUOTE: Elkins-Tanton calls instrument testing 'the icing on the cake.'

Why it matters

This mission could rewrite what we know about planet formation, and the flyby shows how clever engineering can stretch our reach into deep space. For a young person, it's a reminder that math and physics aren't just classroom subjects - they're the tools used to navigate a spacecraft 230 million miles from home, and the same principles that make a

skateboard speed up rolling downhill are tweaked to sling shot a probe across the solar system.

Test what you remember

1. What is the main purpose of the Psyche spacecraft's Mars flyby?
 - (A) To land on Mars and collect samples
 - (B) To take photos of Phobos and Deimos
 - (C) To perform a gravity assist and save propellant
 - (D) To test a new rover design
2. At what speed will Psyche pass by Mars?
 - (A) 5,000 miles per hour
 - (B) 12,333 miles per hour
 - (C) 25,000 miles per hour
 - (D) 50 miles per hour
3. When was the Psyche mission launched?
 - (A) October 2022
 - (B) May 2024
 - (C) October 2023
 - (D) January 2025
4. What will the multispectral imager look for around Mars during the flyby?
 - (A) New moons
 - (B) A faint dusty ring
 - (C) Volcanic eruptions
 - (D) Water ice on the surface
5. Who is the principal investigator of the Psyche mission?
 - (A) Sarah Bairstow
 - (B) Jim Bell
 - (C) Lindy Elkins-Tanton
 - (D) Vera Gorbunova
6. What is the asteroid Psyche believed to be?
 - (A) A captured comet
 - (B) The exposed core of a failed planet
 - (C) A piece of the Moon
 - (D) A solid ball of ice

Answer key (try the quiz first!): 1: C 2: B 3: C 4: B 5: C 6: B

Questions to think about

1. Positive - Scientific and collaborative value: The flyby is a win-win: it saves fuel and allows testing of instruments that will study a unique metal asteroid. International collaboration with ESA and other Mars missions also strengthens scientific networks and cross-calibration, leading to better data for everyone.

2. Negative - Risk and cost concerns: At 12,333 mph, any trajectory error could damage or doom the spacecraft. Skeptics note that the mission costs over \$1 billion, and some argue that funds could be better spent on pressing Earth-bound problems. The Mars flyby also adds mission complexity.

3. **Neutral - Engineering routine with high stakes:** Gravity assists are a standard tool in spaceflight, but each one is uniquely calculated and carries risk. This flyby fits into a long history of planetary slingshots that enabled exploration of the outer solar system; it's an example of how human ingenuity reliably works with physics.

4. Forward-looking - Paving the way for deep-space exploration: Techniques refined here - efficient electric propulsion, precision gravity assists, instrument calibration during flybys - will be crucial for future missions to the icy moons of Jupiter and Saturn, and eventually for human voyages beyond the Earth-Moon system. Psyche's success could directly influence the design of next-generation deep-space probes.

MY THOUGHTS (at least 20 words):